Graphical solutions

1 Consider the function $y = \cos(x - 30)$.

- **a** Sketch the graph of the function for $0 \leq x \leq 360$.
- **b** Sketch the line $y = \frac{1}{2}$ on the same number plane.
- **c** Hence, state all solutions to the equation $\cos(x - 30) = \frac{1}{2}$ over the domain $(0, 360]$.

2 Consider the function $y = \sin(x - 60)$.

- **a** Sketch the graph of the function for $0 \leq x \leq 360$.
- **b** Sketch the line $y = \frac{1}{2}$ on the same number plane.
- **c** Hence, state all solutions to the equation $\sin(x - 60) = \frac{1}{2}$ over the domain $[0, 360)$.

3 Consider the function $y = 2\sin 4x$.

- **a** Sketch the graph of the function for $0 \leq x \leq 120$.
- **b** Sketch the line $y = 1$ on the same number plane.
- **c** Hence, state all solutions to the equation $2\sin 4x = 1$ over the domain $[0, 90]$. Give your answers in degrees.

4 Consider the function $y = 2\sin 2x$.

- **a** Sketch the graph of the function for $0 \leq x \leq 180$.
- **b** State the other function you would add to the graph in order to solve the equation $2\sin 2x = 1$.
- **c** Sketch the graph of this function on the same number plane.
- **d** Hence, state all solutions to the equation $2\sin 2x = 1$ over the domain $[-180^\circ, 180^\circ]$.

5 Consider the function $y = 3\cos 2x + 1$.

- **a** Sketch the graph of the function for $0 \leq x \leq 180$.
- **b** State the other function you would add to the graph in order to solve the equation $3\cos 2x + 1 = \frac{5}{2}$.
- **c** Sketch the graph of this function on the same number plane.
- **d** Hence, state all solutions to the equation $3\cos 2x + 1 = \frac{5}{2}$ over the domain $[0, 180]$.

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6 Consider the function $y = 2\sin 3x - 3$.

a Sketch the graph of the function for $0 \leq x \leq 60$.

b State the other function you would add to the graph in order to solve the equation $2\sin 3x - 3 = -2$.

c Sketch the graph of this fuction on the same number plane.

d Hence, state all solutions to the equation $2\sin 3x - 3 = -2$ over the domain $[0, 60]$.

7 Consider the function $y = -2\cos 3x$.

a Sketch the graph of the function for $0^\circ \leq x \leq 120^\circ$.

b State the other function you would add to the graph in order to solve the equation $-2\cos 3x = -1$.

c Sketch the graph of this function on the same number plane.

d Hence, state all solutions to the equation $-2\cos 3x = -1$ over the domain $[0^\circ, 120^\circ]$.

Exact value equations

8 State whether the following equations have a solution:

a $\cos\theta - 4 = 0$ **b** $9\tan\theta + 4 = 0$

9 State the number of solutions for θ of the following equations in the domain $0^\circ < \theta < 90^\circ$.

a $\cos\theta = -\dfrac{1}{\sqrt{2}}$ **b** $\sin\theta = -\dfrac{\sqrt{3}}{2}$ **c** $\tan\theta = -1$

10 Solve the following equations for $0^\circ \leq \theta \leq 90^\circ$:

a $\sin\theta = \dfrac{1}{\sqrt{2}}$ **b** $\tan\theta = \sqrt{3}$ **c** $\cos\theta = \dfrac{1}{2}$ **d** $\sin\theta = \dfrac{\sqrt{3}}{2}$

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11 Solve the following equations for $0^\circ \leq \theta \leq 360^\circ$:

a $\cos\theta = -\frac{1}{\sqrt{2}}$ **b** $\cos\theta = \frac{1}{2}$

c $\cos\theta = 0$ **d** $\sin\theta = \frac{1}{2}$

e $\sin\theta = 0$ **f** $\sin\theta = -\frac{1}{\sqrt{2}}$

g $\cos\theta = -\frac{1}{\sqrt{2}}$ **h** $\sin\theta = -\frac{\sqrt{3}}{2}$

i $\sin\theta = 1$ **j** $\tan\theta = \sqrt{3}$

k $\tan\theta = 0$ **l** $\tan\theta = -\frac{1}{\sqrt{3}}$

m $4\tan\theta + 2 = -2$ **n** $8\cos\theta - 4 = 0$

o $2\cos\theta + 4 = 3$ **p** $8\sin\theta - 4\sqrt{2} = 0$

Non-exact value equations

12 Solve the following equations for $0^\circ \leq \theta \leq 90^\circ$:

a $\cos\theta = 0.7986$ **b** $\sin\theta = 0.6428$ **c** $\tan\theta = 0.7265$ **d** $\sin\theta = 0.3584$

e $\tan\theta = 2.2460$

13 Solve the following equations for $0^\circ \leq \theta \leq 360^\circ$:

a $\cos\theta = 0.9063$ **b** $\cos\theta = -0.7986$ **c** $\sin\theta = -0.6428$ **d** $\sin\theta = 0.9336$

e $\tan\theta = 0.7002$ **f** $\tan\theta = -0.7265$

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